IX:5
30/7/26

Study of wireshark

AIM:

To study on packet capture tool using wireshark.

Packet Sniffer:

* Sniffs messages being sent/received from/by your computer.
* Store and display the contents of the various protocol fields in the messages.
* passive program
  - → never sends packets itself
  - → no packets addressed to it
  - → receives a copy of all packets (sent/received)

Packet sniffer Structure Diagnostic Tools:

* Tcp dump

  Eg: tcpdump -enx host 10.129.41.2 -w exe 3.out

* wireshark

  Eg: wireshark -rexe 3.out

Description:

* wireshark, a network analysis tool normaly known as the ethernet.

* captures packet in real time and display them in human readable format.

* wireshark includes filters, colour coding and other features that let you dig deep into network traffic.

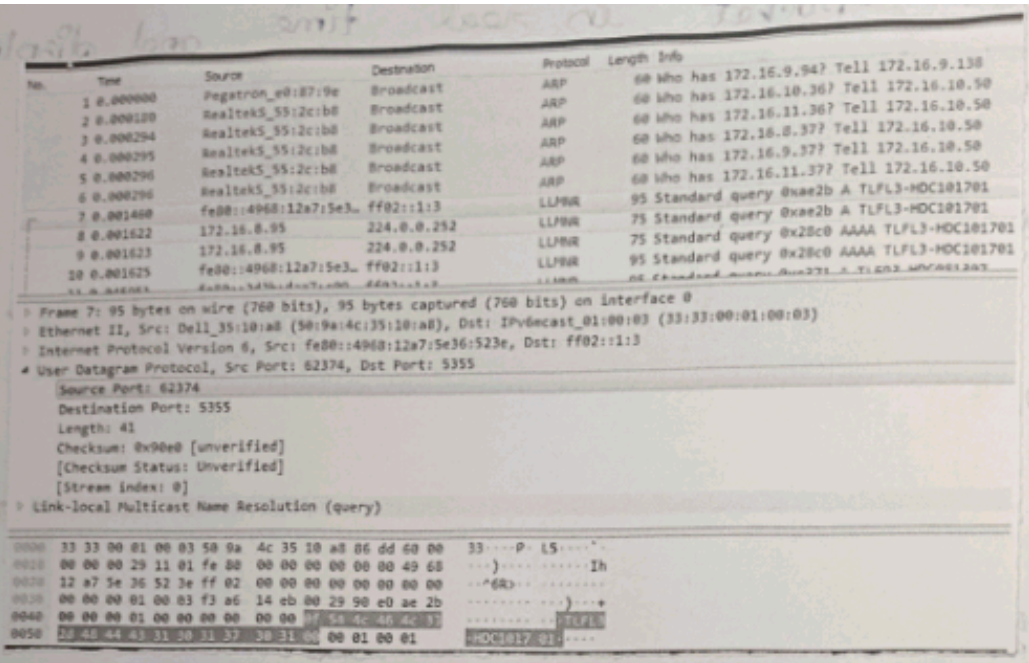
* You can use wireshark to inspect a suspicious program's network traffic.

What we can do with wireshark:

→ capture network traffic
→ Decode packet protocol using dissectors
→ Defines filters - capture and display
→ watch smart statistics
→ Analyze problems
→ Interactivity browse that traffic

wireshark used for:

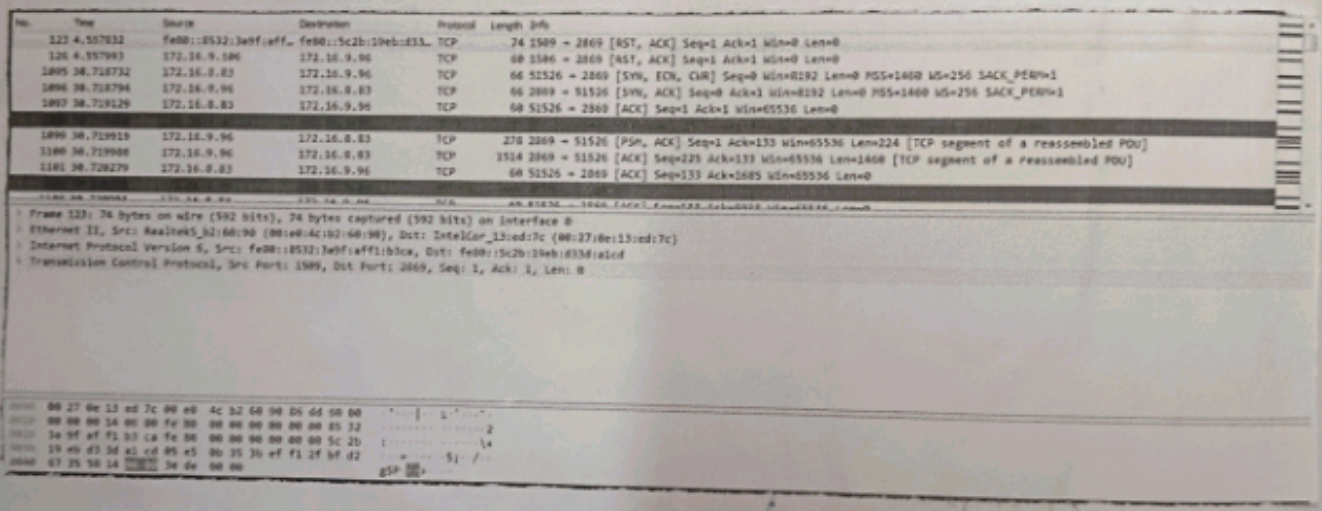
→ Network administrators
→ Network security engineer: examine security problems
→ Debug protocol implementation



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Pegatron_e0:87:9e	Broadcast	ARP	60	Who has 172.16.9.94? Tell 172.16.9.138
2	0.000180	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.10.36? Tell 172.16.10.50
3	0.000294	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.36? Tell 172.16.10.50
4	0.000295	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.8.37? Tell 172.16.10.50
5	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.9.37? Tell 172.16.10.50
6	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.37? Tell 172.16.10.50
7	0.001460	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0xae2b A TLFL3-HOC101701
8	0.001622	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0xae2b A TLFL3-HOC101701
9	0.001623	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0x28c0 AAAA TLFL3-HOC101701
10	0.001625	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0x28c0 AAAA TLFL3-HOC101701

```
▸ Frame 7: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
▸ Ethernet II, Src: Dell_35:10:a8 (50:9a:4c:35:10:a8), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
▸ Internet Protocol Version 6, Src: fe80::4968:12a7:5e36:523e, Dst: ff02::1:3
▾ User Datagram Protocol, Src Port: 62374, Dst Port: 5355
    Source Port: 62374
    Destination Port: 5355
    Length: 41
    Checksum: 0x90e0 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
▸ Link-local Multicast Name Resolution (query)
```

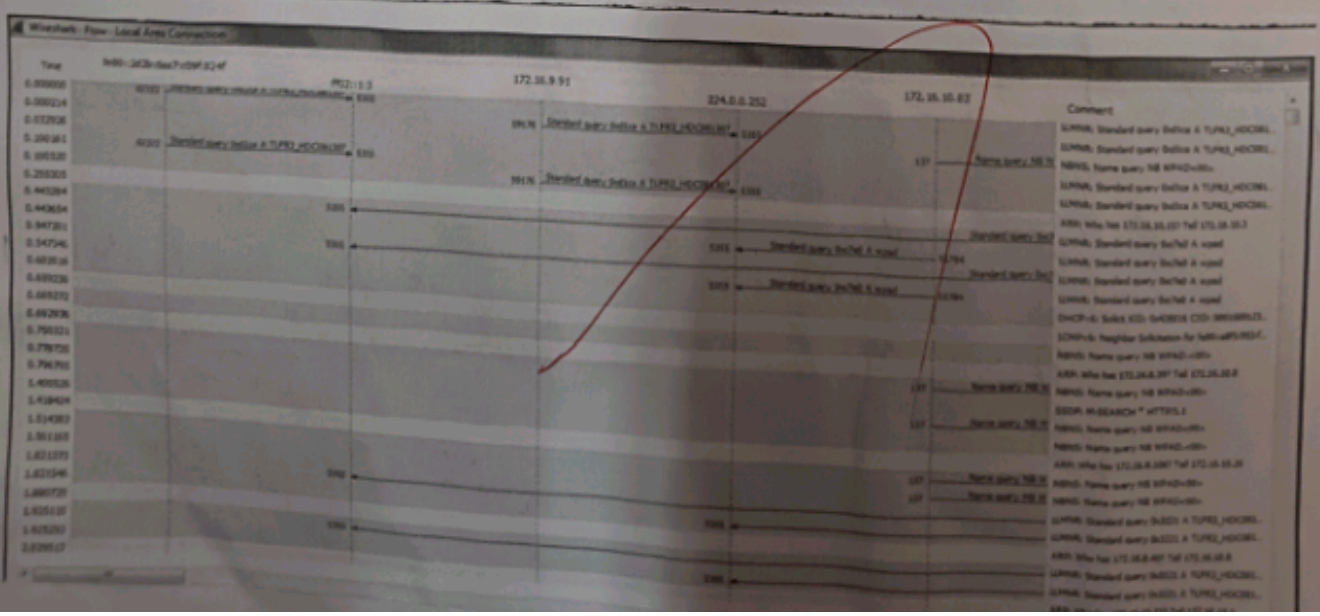
```
0000  33 33 00 01 00 03 50 9a  4c 35 10 a8 86 dd 60 00   33····P· L5····`·
0010  00 00 00 29 11 01 fe 80  00 00 00 00 00 00 49 68   ···)···· ····Ih
0020  12 a7 5e 36 52 3e ff 02  00 00 00 00 00 00 00 00   ··^6R>·· ········
0030  00 00 00 01 00 03 f3 a6  14 eb 00 29 90 e0 ae 2b   ········ ···)···+
0040  00 00 00 01 00 00 00 00  00 00 04 54 4c 46 4c 33   ········ ···TLFL3
0050  2d 48 4f 43 31 30 31 37  30 31 00 00 01 00 01      -HOC1017 01·····
```



No.	Time	Source	Destination	Protocol	Length	Info
123	4.557832	fe80::8532:3a9f:aff…	fe80::5c2b:19eb:d33…	TCP	74	1509 → 2869 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
126	4.557993	172.16.9.106	172.16.9.96	TCP	60	1506 → 2869 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
1095	30.718732	172.16.8.83	172.16.9.96	TCP	66	51526 → 2869 [SYN, ECN, CWR] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
1096	30.718794	172.16.9.96	172.16.8.83	TCP	66	2869 → 51526 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
1097	30.719129	172.16.8.83	172.16.9.96	TCP	60	51526 → 2869 [ACK] Seq=1 Ack=1 Win=65536 Len=0
1098	30.719919	172.16.9.96	172.16.8.83	TCP	278	2869 → 51526 [PSH, ACK] Seq=1 Ack=133 Win=65536 Len=224 [TCP segment of a reassembled PDU]
1100	30.719988	172.16.9.96	172.16.8.83	TCP	1514	2869 → 51526 [ACK] Seq=225 Ack=133 Win=65536 Len=1460 [TCP segment of a reassembled PDU]
1101	30.720279	172.16.8.83	172.16.9.96	TCP	60	51526 → 2869 [ACK] Seq=133 Ack=1685 Win=65536 Len=0

```
▸ Frame 123: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
▸ Ethernet II, Src: RealtekS_b2:60:90 (00:e0:4c:b2:60:90), Dst: IntelCor_13:ed:7c (00:27:0e:13:ed:7c)
▸ Internet Protocol Version 6, Src: fe80::8532:3a9f:aff1:b3ca, Dst: fe80::5c2b:19eb:d33d:a1cd
▸ Transmission Control Protocol, Src Port: 1509, Dst Port: 2869, Seq: 1, Ack: 1, Len: 0
```

```
0000  00 27 0e 13 ed 7c 00 e0  4c b2 60 90 86 dd 60 00   ·'···|·· L·`···`·
0010  00 00 00 14 06 80 fe 80  00 00 00 00 00 00 85 32   ········ ·······2
0020  3a 9f af f1 b3 ca fe 80  00 00 00 00 00 00 5c 2b   :····· ·····\+
0030  19 eb d3 3d a1 cd 05 e5  0b 35 3b ef f1 2f bf d2   ···=···· ·5;··/··
0040  67 35 50 14 [illegible] 3e de  00 00              g5P· ··>·
```



Getting Wireshark:

→ Wireshark can be downloaded for window

→ For Linux or another UNIX-like system

→ Found in it package repositories

Capturing And Analysing Packets using Wireshark Tool:

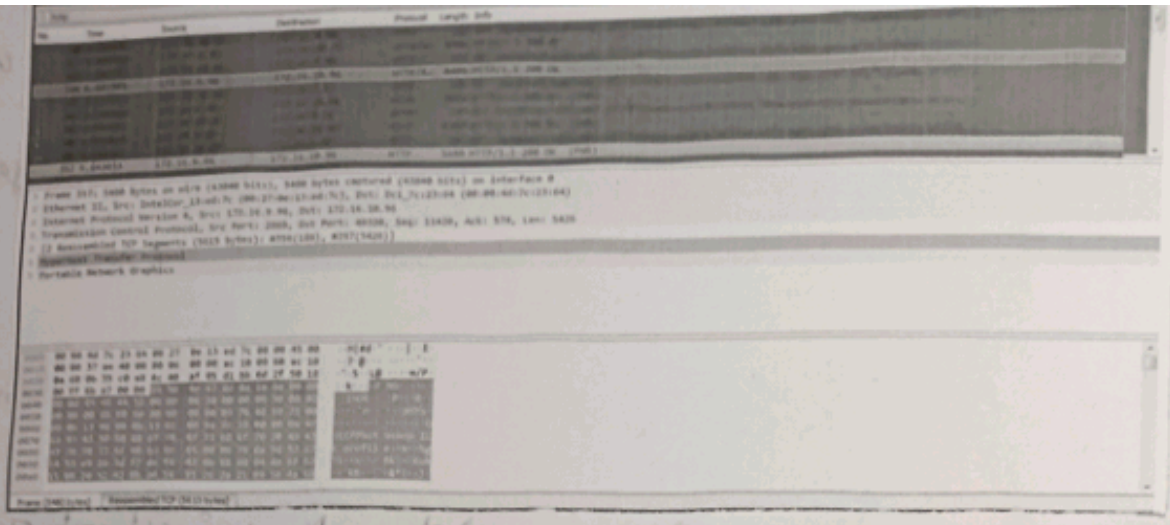
To filter, capture, view, packets in wireshark Tool. Capture 100 packets from the Ethernet: IEEE 802.3 LAN Interface and saved it.

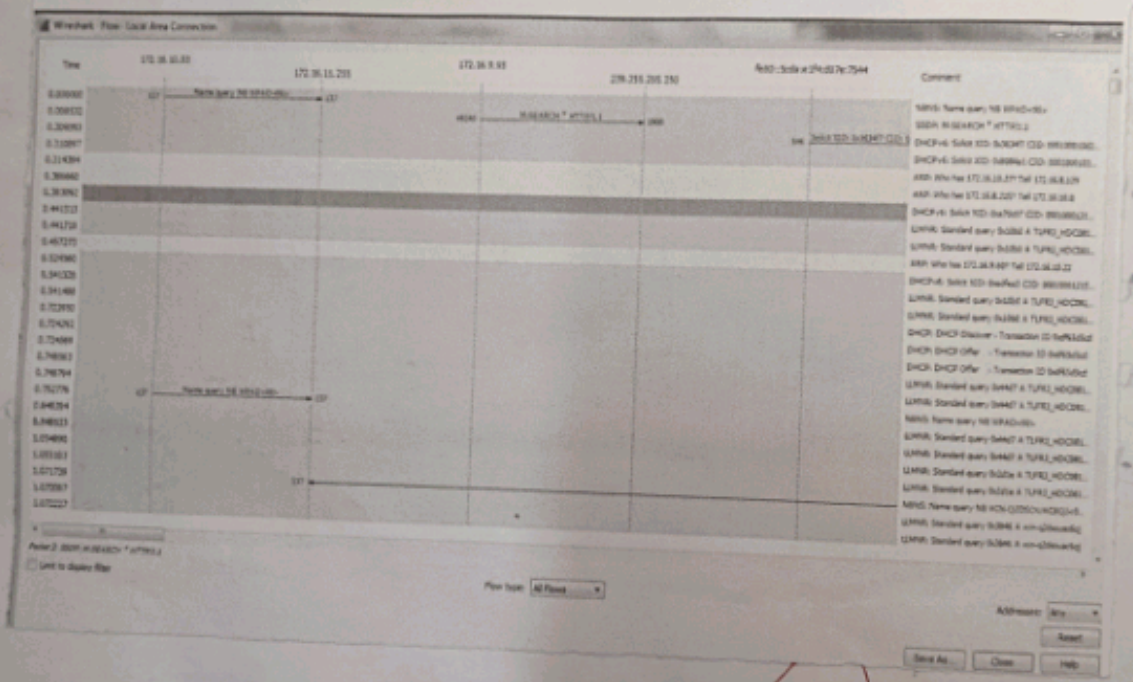
PROCEDURE:

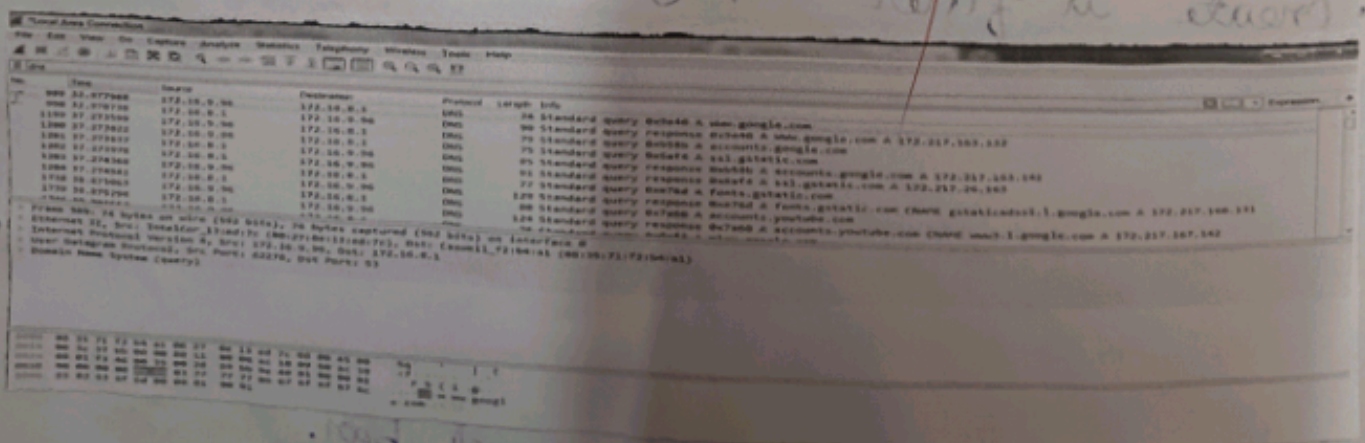
* Select Local Area Connection in wireshark
* Go to capture option
* Select stop capture automatically after 100 packets
* Then click start capture.
* Save the packets.

1. Create a filter display only TCP / UDP

* Select Local Area Connection in wireshark.
* Go to capture option
* Select stop capture automatically after 100 packets
* Then click start capture.
* Search TCP packets in search bar.
* To see flow graph click statistics flow graph.
* Save the packets.







2. Display a filter to display only ARP packets and inspect

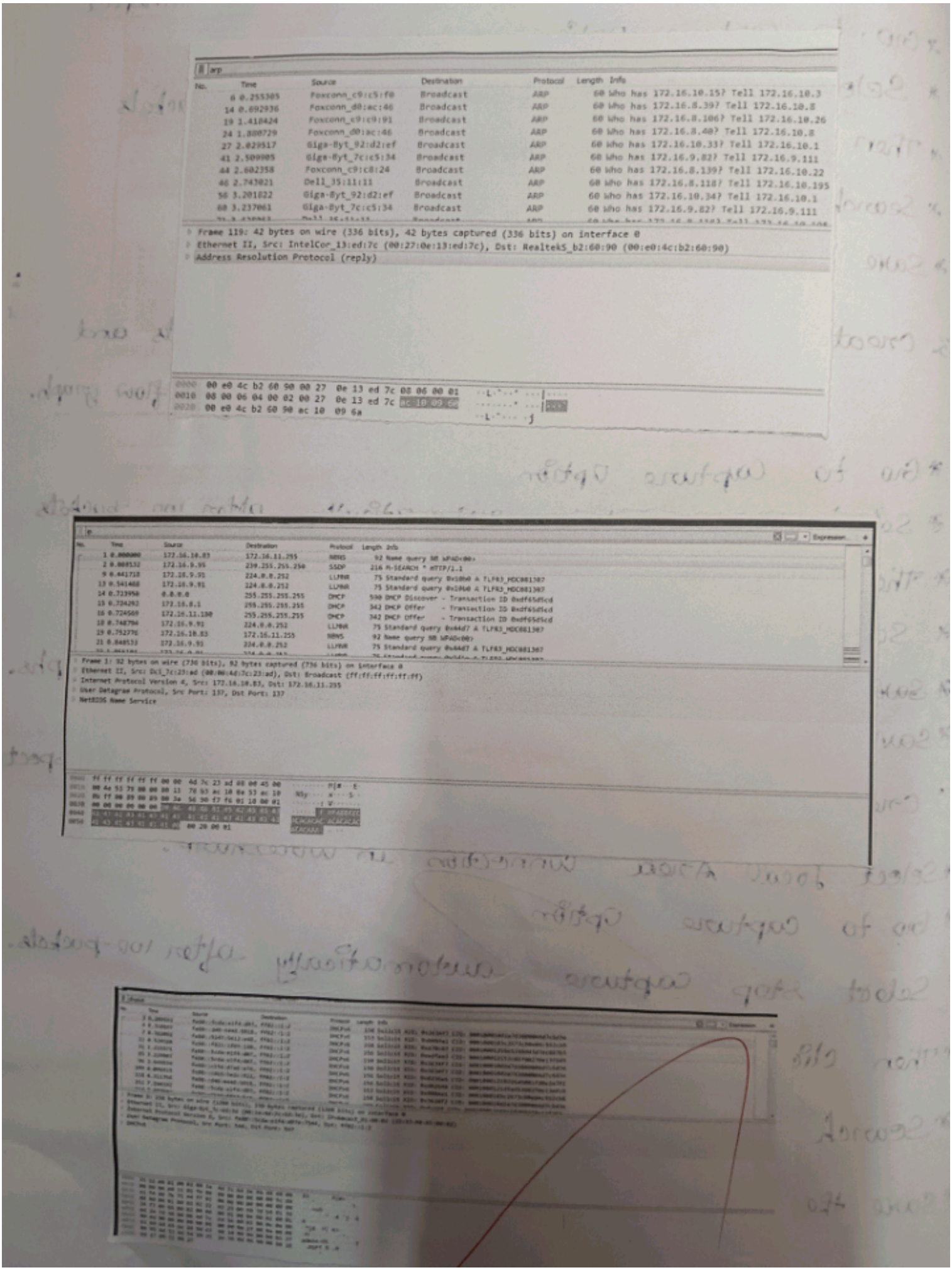
* Go to capture Option
* Select stop automatically after 100 packets
* Then click start capture.
* Search ARP packets in search bar
* Save the packets.

3. Create a filter to display only DNS packets and provide flow graph.

* Go to capture Option
* Select stop capture automatically after 100 packets
* Then click start capture.
* Search DNS packets in search bar
* To see flow graph click statistics flow graphs.
* Save the packets.

4. Create a filter to display only HTTP packets and inspect

* Select local Area connection in wireshark.
* Go to capture Option
* Select stop capture automatically after 100 packets.
* Then click start capture.
* Search HTTP packets in search bar.
* Save the packets.


arp
No. Time Source Destination Protocol Length Info
6 0.255305 Foxconn_c9:c5:f0 Broadcast ARP 60 Who has 172.16.10.15? Tell 172.16.10.3
14 0.692936 Foxconn_d0:ac:46 Broadcast ARP 60 Who has 172.16.8.39? Tell 172.16.10.8
19 1.410424 Foxconn_c9:c9:91 Broadcast ARP 60 Who has 172.16.8.106? Tell 172.16.10.26
24 1.808729 Foxconn_d0:ac:46 Broadcast ARP 60 Who has 172.16.8.40? Tell 172.16.10.8
27 2.029517 Giga-Byt_92:d2:ef Broadcast ARP 60 Who has 172.16.10.33? Tell 172.16.10.1
41 2.509005 Giga-Byt_7c:c5:34 Broadcast ARP 60 Who has 172.16.9.82? Tell 172.16.9.111
44 2.602358 Foxconn_c9:c8:24 Broadcast ARP 60 Who has 172.16.8.139? Tell 172.16.10.22
46 2.743021 Dell_35:11:11 Broadcast ARP 60 Who has 172.16.8.118? Tell 172.16.10.195
56 3.201822 Giga-Byt_92:d2:ef Broadcast ARP 60 Who has 172.16.10.34? Tell 172.16.10.1
60 3.237061 Giga-Byt_7c:c5:34 Broadcast ARP 60 Who has 172.16.9.82? Tell 172.16.9.111
Frame 119: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
Ethernet II, Src: IntelCor_13:ed:7c (00:27:0e:13:ed:7c), Dst: Realtek5_b2:60:90 (00:e0:4c:b2:60:90)
Address Resolution Protocol (reply)
0000 00 e0 4c b2 60 90 00 27 0e 13 ed 7c 08 06 00 01
0010 08 00 06 04 00 02 00 27 0e 13 ed 7c ac 10 09 6c
0020 00 e0 4c b2 60 90 ac 10 09 6a

No. Time Source Destination Protocol Length Info
1 0.000000 172.16.10.83 172.16.11.255 NBNS 92 Name query NB WPAD<00>
2 0.000532 172.16.9.95 239.255.255.250 SSDP 216 M-SEARCH * HTTP/1.1
9 0.441718 172.16.9.91 224.0.0.252 LLMNR 75 Standard query 0x1060 A TLF83_MDC881307
13 0.541468 172.16.9.91 224.0.0.252 LLMNR 75 Standard query 0x1060 A TLF83_MDC881307
14 0.723950 0.0.0.0 255.255.255.255 DHCP 590 DHCP Discover - Transaction ID 0xdf65d5cd
15 0.724292 172.16.8.1 255.255.255.255 DHCP 342 DHCP Offer - Transaction ID 0xdf65d5cd
16 0.724569 172.16.11.130 255.255.255.255 DHCP 342 DHCP Offer - Transaction ID 0xdf65d5cd
18 0.748794 172.16.9.91 224.0.0.252 LLMNR 75 Standard query 0x44d7 A TLF83_MDC881307
19 0.752776 172.16.10.83 172.16.11.255 NBNS 92 Name query NB WPAD<00>
21 0.848533 172.16.9.91 224.0.0.252 LLMNR 75 Standard query 0x44d7 A TLF83_MDC881307
Frame 1: 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface 0
Ethernet II, Src: Dell_7c:23:ad (00:00:4d:7c:23:ad), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 172.16.10.83, Dst: 172.16.11.255
User Datagram Protocol, Src Port: 137, Dst Port: 137
NetBIOS Name Service

5. Creates a filter to display only IP/ICMP packets and inspect

* Select local Area Connection in Wireshark.
* Go to Capture Option
* Select stop Capture automatically after 100 packets.
* Then click Start Capture.
* Search ICMP/ IP packets in search bar.
* Save the packets

6. Creates a ~~packets~~ Filters to display only DHCP packets & inspect.

* Select local Area Connection in wireshark.
* Go to capture option
* Select stop capture automatically after 100 packet.
* Then click start packets.
* Search DHCP packets in search bar.
* Save the packets.

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RESULT:

Bully 18 Thus, The study of wireshark is learned successfully.